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EE202 Lab #6

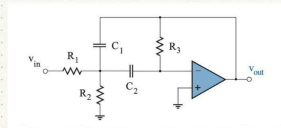
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Introduction:

In this lab we design, build, and test a Rausch active band-pass filter using an LM741 operational amplifier. We start by deriving the filter's transfer function in standard band-pass form and identifying the midband gain K , the center angular frequency ω_0 , and the quality factor Q . Using these relationships, we choose resistor values for a target center frequency $f_0 = 1$ kHz, $Q = 5$, and $|K| = 2$, with capacitors fixed at $C = 10$ nF. After completing the design, we build the circuit with available resistors whose measured values closely match the calculated ones, then experimentally determine the actual midband gain, center frequency, and bandwidth. From the measured bandwidth we compute $Q = f_0 / B$ and compare the experimental results with the theoretical design values to evaluate the performance and accuracy of the band-pass filter.

Test Setup and Equation Derivation:



Need to use KCL + Ideal op-amp property

$$H(s) = \frac{V_{out}}{V_{in}}, \quad V_f = \text{ground} = 0, \quad V_- = 0$$

- KCL at the inverting input

i_{C2} through C_2 equal i_{R3} through R_3

$$\frac{V_x - 0}{1/s_2} + \frac{V_{out} - 0}{R_3} = 0 \rightarrow s_2 V_x + \frac{V_{out}}{R_3} = 0$$

$$V_x = -\frac{V_{out}}{s_2 R_3} \quad \downarrow$$

$$\frac{V_x - V_{in}}{R_1} + \frac{V_x}{R_2} + (V_x - V_{out})s_1 + (V_x - 0)s_2 = 0$$

$$V_x \left(\frac{1}{R_1} + \frac{1}{R_2} + s_1 + s_2 \right) = \frac{V_{in}}{R_1} + s_1 V_{out}$$

substitute!

$$\left(-\frac{V_{out}}{s_2 R_3} \right) \left(\frac{1}{R_1} + \frac{1}{R_2} + s_1 + s_2 \right) = \frac{V_{in}}{R_1} + s_1 V_{out}$$

$$V_{out} \left(\frac{1}{R_1 R_2} + s_1 + s_2 \right) = -s_2 R_3 \left(\frac{V_{in}}{R_1} \right) - s_1 s_2 R_3 V_{out}$$

$$V_{out} \left(s_1^2 C_1 C_2 R_3 + s_1 (C_1 + C_2) + \frac{1}{R_1 R_2} \right) = -V_{in} \left(\frac{s_2 R_3}{R_1} \right)$$

$$H(s) = \frac{-\frac{s_2 R_3}{R_1}}{s_1^2 C_1 C_2 R_3 + s_1 (C_1 + C_2) + \frac{1}{R_1 R_2}}$$

normalizing: $\downarrow$

$$\text{numerator: } \frac{-\frac{s_2 R_3}{R_1}}{C_1 C_2 R_3} = \frac{-s}{R_1 C_1}$$

denominator:

$$s^2 + s \frac{C_1 + C_2}{C_1 C_2 R_3} + \frac{1}{(R_1 R_2) R_3 C_1 C_2}$$

Combined transfer function:

$$H(s) = \frac{-s}{R_1 C_1} \cdot \frac{1}{s^2 + s \frac{C_1 + C_2}{C_1 C_2 R_3} + \frac{1}{(R_1 || R_2) R_3 C_1 C_2}}$$

Convert to target form: $H(s) = \frac{K s \frac{\omega_0}{Q}}{s^2 + s \frac{\omega_0}{Q} + \omega_0^2}$

$$\omega_0^2 = \frac{1}{(R_1 || R_2) R_3 C_1 C_2}$$

$$\omega_0 = \frac{1}{\sqrt{(R_1 || R_2) R_3 C_1 C_2}}$$

$$\frac{\omega_0}{Q} = \frac{C_1 + C_2}{C_1 C_2 R_3} \rightarrow Q = \omega_0 \left(\frac{C_1 C_2 R_3}{C_1 + C_2} \right)$$

$$Q = \frac{1}{\sqrt{(R_1 || R_2) R_3 C_1 C_2}} \cdot \frac{C_1 C_2 R_3}{C_1 + C_2}$$

$$= \frac{\sqrt{R_3}}{\sqrt{R_1 || R_2}} \cdot \frac{\sqrt{C_1 C_2}}{C_1 + C_2}$$

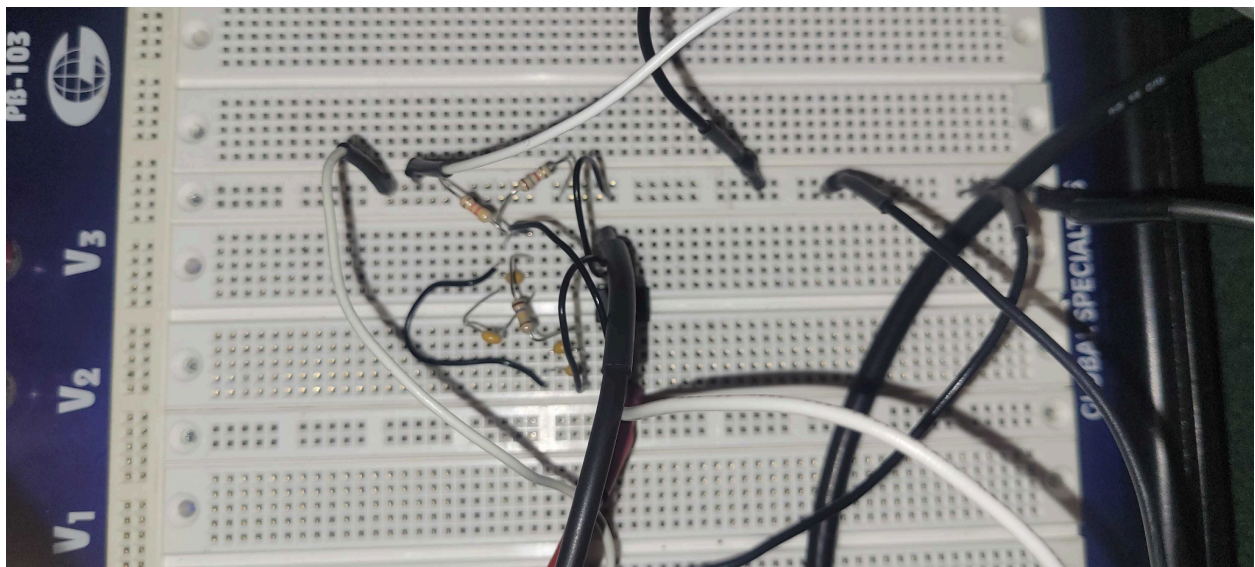
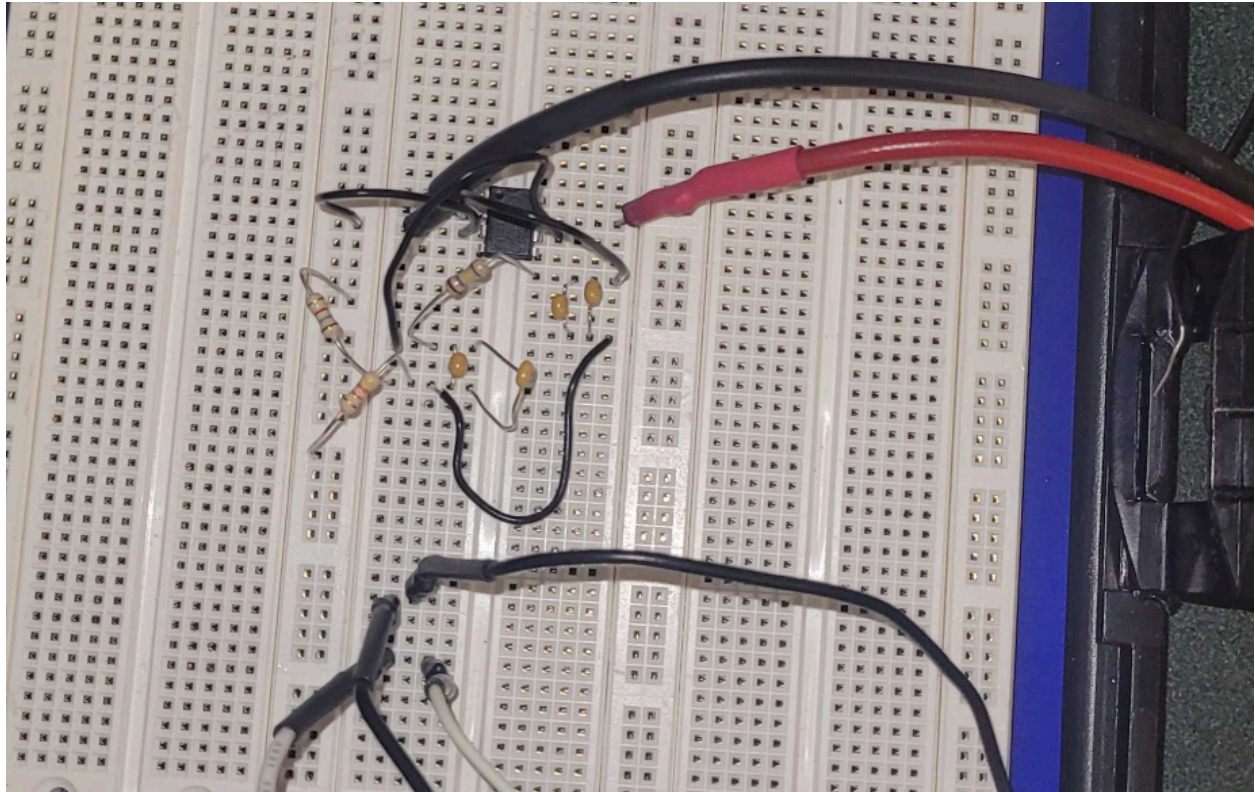
$$= \frac{\sqrt{\frac{R_3}{R_1 || R_2}}}{\sqrt{\frac{C_2}{C_1} + \sqrt{\frac{C_2}{C_1}}}} \quad \checkmark$$

$$K \left(s \frac{\omega_0}{Q} \right) = \frac{-s}{R_1 C_1}$$

$$K \left(\frac{C_1 + C_2}{C_1 C_2 R_3} \right) = -\frac{1}{R_1 C_1}$$

$$K = -\frac{1}{R_1 C_1} \cdot \frac{C_1 C_2 R_3}{C_1 + C_2}$$

$$K = -\frac{C_2 R_3}{R_1 (C_1 + C_2)} = -\left(\frac{R_3}{R_1} \right) \left(\frac{C_2}{C_1 + C_2} \right)$$



Measurements:

Frequency (Hz)	Vin (mV)	Vout (mV)
100	100	4.14

300	100	13.29
500	100	26.38
800	100	78.79
900	100	130.51
950	100	168.72
1000	100	196.55
1050	100	186.65
1100	100	153.84
1200	100	102.18
1500	100	48.91
2000	100	27.56
3000	100	15.62
5000	100	8.51
10000	100	4.24

Discussion of Results:

Analysis and Discussion:

Output voltage peaked at 1000 Hz with a value of 196.55mV.

$f_{o, \text{measured}} = 1000 \text{ Hz}$

Gain at the center frequency:

$|K|_{\text{measured}} = V_{\text{peak}} / V_{\text{in}}$

$|K|_{\text{measured}} = 196.55 \text{ mV} / 100 \text{ mV}$

$|K|_{\text{measured}} = 1.97$

Percent error of 1.5% with target of $|K| = 2$.

The bandwidth was determined by identifying the -3dB frequencies (f_L and f_H) where the output voltage drops to 0.707 of the peak value.

Calculation of -3dB Voltage Level:

$V_{3\text{dB}} = V_{\text{peak}} * 0.707$

$V_{3\text{dB}} = 196.55 \text{ mV} * 0.707 = 138.96 \text{ mV}$

Lower Cutoff Frequency (f_L):

The voltage crosses 138.96 mV between 900 Hz (130.51 mV) and 950 Hz (168.72 mV).

Linear interpolation was used to approximate the exact frequency:

$f_L = 900 + [(138.96 - 130.51) / (168.72 - 130.51)] * (950 - 900)$

$$f_L = 900 + (8.45 / 38.21) * 50$$

$$f_L = 911.1 \text{ Hz}$$

Upper Cutoff Frequency (f_H):

The voltage crosses 138.96 mV between 1100 Hz (153.84 mV) and 1200 Hz (102.18 mV).

Using linear interpolation:

$$f_H = 1100 + [(138.96 - 153.84) / (102.18 - 153.84)] * (1200 - 1100)$$

$$f_H = 1100 + (-14.88 / -51.66) * 100$$

$$f_H = 1128.8 \text{ Hz}$$

Experimental Bandwidth (B):

$$B_{\text{measured}} = f_H - f_L$$

$$B_{\text{measured}} = 1128.8 \text{ Hz} - 911.1 \text{ Hz}$$

$$B_{\text{measured}} = 217.7 \text{ Hz}$$

Experimental Quality Factor (Q):

$$Q_{\text{measured}} = f_o / B$$

$$Q_{\text{measured}} = 1000 \text{ Hz} / 217.7 \text{ Hz}$$

$$Q_{\text{measured}} = 4.59$$

Center Frequency (f_o) 1 kHz vs 1 kHz, Error 0%

Gain ($|K|$) 2.0 vs 1.97, Error -1.5%

Bandwidth (B) 200 Hz 217.7 Hz, Error +8.85%

Quality Factor (Q) 5.0 4.59, Error -8.2%

The circuit performed consistently with the design specifications. The center frequency was identified exactly at 1 kHz. The measured gain of 1.97 was extremely close to the target of 2. The Quality Factor (Q) was measured at 4.59, which is slightly lower than the target of 5. This broadening of the bandwidth (from the ideal 200 Hz to 217.7 Hz) is likely due to the tolerance of the physical capacitors and resistors used in the construction, particularly the sensitivity of Q to variations in the feedback resistance R3.

Conclusion:

From this lab, we learned how a Rausch active band pass filter built around an op amp can be designed to meet a target center frequency, gain, and quality factor. By selecting resistor values from the design equations, building the circuit, and sweeping the input frequency, we measured the actual midband gain, center frequency, and bandwidth and compared them to the design targets. The results showed that the filter behaved as a narrowband amplifier around 1 kHz and mostly matched the expected response, while small shifts in gain and bandwidth highlighted how component tolerances and op amp nonidealities affect the performance of high Q filters.

Peter built the breadboard/filter and logged the data.

John helped with picking correct pieces for equivalent resistance and capacitance and did the bulk of the data analysis.

We both worked on the write up together.